# Running V FOODS Locally

A practical step-by-step setup guide for running the V FOODS frontend and backend on your local computer for development and testing.

---

## 1. Prerequisites

Before starting, make sure you have the following installed on your computer:

- **Node.js**: Version 18 or newer (comes with `npm`).
- **Python**: Version 3.10 or newer (comes with `venv` and `pip`).
- **Git**: For pulling the code and managing branches.

---

## 2. External Service Dependencies

V FOODS connects to three cloud services. To run the full system with live data, you will need credentials from each:

1. **Supabase Project**: Provides the PostgreSQL database, authentication, Row-Level Security, and Realtime WebSocket events.
2. **Paytm Merchant Credentials**: Provides sandbox/staging gateway credentials ( `PAYTM_MID` , `PAYTM_MERCHANT_KEY` , `PAYTM_WEBSITE` , `PAYTM_CHANNEL_ID` , `PAYTM_INDUSTRY_TYPE_ID` ) for testing wallet top-ups and direct order checkout with mock UPI/Cards.
3. **MSG91 Account (Optional in Local Dev)**: Provides SMS and WhatsApp messaging APIs for order ready notifications. If left blank, the app logs notifications to the console instead of sending live SMS.

---

## 3. Environment Variables Setup

Environment variables are configuration settings that tell the app where to connect without exposing private passwords in the code.

### Frontend Configuration

Create a file named `.env` inside the `frontend/` directory with these variable names:

- `VITE_SUPABASE_URL` : The web address of your Supabase project (e.g. `https://xyzcompany.supabase.co` ).
- `VITE_SUPABASE_ANON_KEY` : The public anonymous API key provided in your Supabase project settings.

- `VITE_API_URL`: The local web address of your running FastAPI backend (typically `http://localhost:8000`).

**Backend Configuration**

Create a file named `.env` inside the `backend/` directory with these variable names:

- `SUPABASE_URL`: The web address of your Supabase project.
- `SUPABASE_SERVICE_ROLE_KEY`: The private service role key from Supabase (used by the backend for admin operations).
- `PAYTM_MID`: Your Paytm Merchant ID (use staging MID for testing).
- `PAYTM_MERCHANT_KEY`: The private cryptographic AES/HMAC merchant key provided by Paytm.
- `PAYTM_WEBSITE`: `WEBSTAGING` (for test mode) or `DEFAULT` (for production).
- `PAYTM_CHANNEL_ID`: `WAP` (for mobile web checkout) or `WEB`.
- `PAYTM_INDUSTRY_TYPE_ID`: `Retail`.
- `PAYTM_ENV`: `STAGE` (for sandbox testing) or `PROD` (for live).
- `WEBHOOK_BASE_URL`: The public HTTPS address where Paytm sends webhook callbacks (e.g. Ngrok tunnel or cloud host URL).
- `FRONTEND_URL`: The address of the frontend app (e.g. `http://localhost:5173`).
- `MSG91_AUTH_KEY`: Your MSG91 authorization key for messaging (optional in dev).
- `MSG91_WHATSAPP_INTEGRATED_NUMBER`: The verified WhatsApp sender number registered on MSG91.
- `QR_SECRET`: A random secret passphrase used to generate secure HMAC-signed QR codes for order collection.

---

## 4. Starting the Frontend (React + Vite)

1. Open your terminal and navigate to the `frontend` folder: `bash cd frontend`
2. Install the necessary project dependencies: `bash npm install`
3. Start the local development server: `bash npm run dev`
4. The website will open in your browser at `http://localhost:5173`. You can log in using test credentials or create a new student account.

---

## 5. Starting the Backend (FastAPI Python)

1. Open a second terminal window and navigate to the `backend` folder: `bash cd backend`
2. Create and activate a Python virtual environment:
3. On Windows: `bash python -m venv venv .\venv\Scripts\activate`
4. On macOS / Linux: `bash python3 -m venv venv source venv/bin/activate`
5. Install required Python packages: `bash pip install -r requirements.txt`

6. Start the FastAPI server: `bash uvicorn main:app --reload --port 8000`
7. The API will start at `http://localhost:8000`. You can test endpoints directly in your browser using the built-in documentation at `http://localhost:8000/docs`.

---

## 6. Database Migrations Setup

1. Log in to your Supabase web dashboard.
2. Open the **SQL Editor**.
3. Run the SQL migration scripts located in the `supabase/migrations/` folder in sequential order ( `001_...`, `002_...`, etc.).
4. This will create all required tables ( `outlets`, `menu_items`, `orders`, `wallets`, etc.), configure Row-Level Security rules, and install the atomic wallet procedures.

---